

12.12 Solving One-Step Multiplication and Division Equations Algebraically

Lesson Introduction

 Benchmark	MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.		 Learning Target	I can solve one-step multiplication and division equations in one variable algebraically.
 Benchmark Coherence	Building On MA.5.AR.2.4 Given a mathematical or real-world context, write an equation involving any of the four operations to determine the unknown whole number with the unknown in any position.		Working Toward MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.	
 Essential Questions	How can you use inverse operations to isolate the variable to solve one-step equations algebraically?			
 Lesson Narrative	In this lesson, students will solve multiplication and division one-step equations algebraically using inverse operations. Students begin this lesson with an activity that allows them to explore how performing inverse operations when trying to find the solution of a variable in an equation helps to isolate the variable and keep the equation balanced. Students then use inverse operations algebraically to isolate the variable and find a solution.			
 Component Summary	12.12.1 Warm-Up (~5 min.)	Write equations with a given solution (<i>SE p. 346</i>)	12.12.3 Guided Practice (10-15 min.)	Collaboratively practice solving one-step multiplication and division equations algebraically (<i>SE p. 348</i>)
	12.12.2 Guided Instruction (10-15 min.)	Learn how to solve equations using inverse operations (<i>SE p. 347</i>)	12.12.4 Your Turn! (10-15 min.)	Independently practice solving one-step multiplication and division equations algebraically (<i>SE p. 349</i>)
	12.12.5 Wrap-Up (~5 min.)	Explain and correct an error in a sample of student work (<i>SE p. 350</i>)		
 Resources	Glossary		Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		Consider creating an anchor chart with instructions, examples, and visual models.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may believe that since $7x$ divided by 7 results in x that the coefficient of x is zero instead of one. - Students may not perform the inverse operation to both sides of the equation. - Students may not know what to do when the variable is on the right side of the equal sign.	- During 12.12.2 Guided Instruction, it is important to link the algebraic methods to the visual and concrete methods previously used. - During 12.12.2 Guided Instruction, explain what the equal sign means to the equation and how it keeps both sides of the equation balanced. - For question 4 in 12.12.2 Guided Instruction, tell students that the variable can be on either side of the equation. If they prefer, demonstrate how to rewrite the equation with the variable on the left side. Explain that this is the symmetric property of equality, if $a = b$ then $b = a$.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Take Turns In 12.12.3 Guided Practice, students explain the rationale for their solutions and listen to their partner's rationales. Taking turns deciding, explaining, and listening limits one student from taking over and the other not participating. Taking turns can also give students more opportunities to construct logical arguments and critique others' reasonings.	MA.K12.MTR.4.1: Discussion and Reflection In the 12.12.5 Wrap-Up, students identify and correct errors in a sample of student work. Students write an explanation of the error and describe how they would correct it. This activity will help students avoid making similar errors themselves in the future.
 Differentiate Instruction	12.12.4 Your Turn! involves students working independently. If students struggle to solve equations algebraically, provide them with concrete manipulatives or tell them to draw visual models. Also, consider creating and displaying an anchor chart with instructions, examples, and visual models to reference. Finally, if students struggle with their multiplication and division facts, allow them to use a multiplication chart until they become fluent in these facts.	
Lesson Notes:		

12.12.1 Warm-Up: Different Problems, Same Solution

Component Overview: Students write four equations that all have a solution of $x = 3$. The directions do not specify what type of equations students should write. Clarify that students should write one-step multiplication and division equations. Consider adding additional requirements, such as two multiplication and two division equations or requiring each equation to include a negative integer.



12.12.1 Warm-Up: Different Problems, Same Solution

1. Write four equations whose solutions are all $x = 3$.

Answers will vary. $6x = 18$; $4x = 12$; $12x = 36$; $\frac{x}{3} = 1$



If students finish early, challenge them to write a real-world problem that can be modeled by each of their equations.

12.12.2 Guided Instruction: Using Inverses to Solve Equations

Component Overview: Students work with a partner to solve equations using inverse operations. Consider creating an anchor chart and displaying it for students to reference in subsequent components. Explain the connections between using inverses and using visual models.



12.12.2 Guided Instruction: Using Inverses to Solve Equations

Inverse operations are used to solve equations by isolating a variable of interest.

When solving one-step equations using visual models, the inverse operation performed on one side of the model is also performed on the other side to keep the equation balanced.

1. With your partner, re-visit the meaning of balanced equations. Write a brief summary explaining what it means for equations to be balanced and how balance is maintained when solving. Include examples as part of your summary.

Answers will vary. For an equation to remain balanced, inverse operations will need to be performed on both sides of the equal sign.

2. Using the definition of inverse operations, complete the table to indicate which operation would reverse the given operation.

Operation	Inverse Operation
Multiplication ($\times$)	<i>Division ($\div$)</i>
Division ($\div$)	<i>Multiplication ($\times$)</i>



Explain the meaning of the equal sign and how it keeps both sides of the equation balanced. Remind students of the balance scale they saw in previous lessons. Explicitly link the algebraic methods to the visual and concrete methods previously used.

12.12.2 Guided Instruction: Using Inverses to Solve Equations (continued)

3. Consider the equation $7m = 21$ to complete the following.

a. Discuss with your partner how to solve $7m = 21$.
Answers will vary. In the equation, $7m$ means 7 times m . The inverse of multiplication is division, so you should divide both sides of the equation by 7 to solve for m .

b. What operation is occurring between the variable m and its coefficient of 7?
Multiplication

c. To isolate the variable m from its coefficient of 7, what inverse operation should be performed?
Division

d. Work with your partner to determine the value of m .
 $m = 3$

4. Consider the equation $13 = \frac{h}{-5}$ to complete the following:

a. What mathematical operation is represented by $\frac{h}{-5}$?
Division

b. What is the inverse operation?
Multiplication

c. Perform the inverse operation in the boxes below.

$$\boxed{-5} (13) = \left(\frac{h}{-5}\right) \boxed{(-5)}$$

d. What is the value of h ? *$h = -65$*

e. Discuss with your partner how to check whether or not your solution is correct. Then, check your solution. *Answers will vary. Substitute -65 in for h . Divide -65 by -5 and the result will equal 13.*



After solving for m , facilitate a discussion regarding what happened when the coefficient 7 was divided by 7.

It's important to distinguish that the coefficient of m becomes 1 in this case, not 0. After solving addition and subtraction equations, students come to understand that the constant term is eliminated when zero pairs are made. The term being added or subtracted becomes 0 when solving.

For multiplication and division, this is not the case. It's important to make this distinction and develop this understanding in 6th grade when students are working with one-step equations to develop strong foundations as students become more flexible in future courses where they'll learn to solve two-step, multistep, and literal equations.



To encourage mathematical reasoning, ask questions like:

- *Does this answer make sense?*
- *How did you decide the answer made sense?*
- *How can you determine if the answer will be positive or negative?*

12.12.3 Guided Practice: Solving One-Step Multiplication and Division Equations Algebraically

Component Overview: Students collaboratively practice solving one-step multiplication and division equations algebraically. Then students use substitution to check their solutions. Circulate while students work to monitor progress and decide which students you will call on to share their solutions during the whole-class debrief.



12.12.3 Guided Practice: Solving One-Step Multiplication and Division Equations Algebraically

1. Complete the table.

Equation	Description of How to Solve	Work	Solution
$6x = 18$	Divide both sides of the equation by 6.	$\frac{6x}{6} = \frac{18}{6}$	$x = 3$
$\frac{h}{5} = 7$	Multiply both sides of the equation by 5.	$(5)\frac{h}{5} = 7(5)$	$h = 35$
$-36 = \frac{m}{-6}$	Multiply both sides of the equation by -6.	$(-6)(-36) = \left(\frac{m}{-6}\right)(-6)$	$m = 216$
$-32 = 8y$	Divide both sides of the equation by 8.	$\frac{-32}{8} = \frac{8y}{8}$	$y = -4$
$4 = -\frac{b}{12}$	Multiply both sides of the equation by -12.	$(-12)(4) = \left(-\frac{b}{12}\right)(-12)$	$b = -48$



Take Turns

Students explain the rationale for their solutions and listen to their partner's rationale. Taking turns deciding, explaining, and listening limits one student from taking over and the other not participating. Taking turns can also give students more opportunities to construct logical arguments and critique others' reasonings.

12.12.3 Guided Practice: Solving One-Step Multiplication and Division Equations Algebraically (continued)

2. With your partner, verify that each solution is correct using substitution.

$$\begin{aligned}(6)(3) &= 18 \\ 18 &= 18\end{aligned}$$

$$\begin{aligned}\frac{35}{5} &= 7 \\ 7 &= 7\end{aligned}$$

$$\begin{aligned}-36 &= \frac{216}{-6} \\ -36 &= -36\end{aligned}$$

$$\begin{aligned}-32 &= 8(-4) \\ -32 &= -32\end{aligned}$$

$$\begin{aligned}4 &= -\frac{-48}{12} \\ 4 &= 4\end{aligned}$$



The partner discussions about the correctness of their solutions provide an auditory entry point for students to discuss their methods and solutions.

12.12.4 Your Turn!

Component Overview: Students independently practice solving one-step multiplication and division equations algebraically. Students may find it helpful to reference an anchor chart with instructions, examples, and visual models.



12.12.4 Your Turn!

Solve each equation algebraically. Show your work.

1. $\frac{a}{4} = 10$ $\frac{a}{4}(4) = 10(4)$ $a = 40$	2. $-11 = \frac{x}{8}$ $(8)(-11) = \frac{x}{8}(8)$ $x = -88$
3. $-12p = -84$ $\frac{-12p}{-12} = \frac{-84}{-12}$ $p = 7$	4. $-42 = 6x$ $\frac{-42}{6} = \frac{6x}{6}$ $x = -7$
5. $18c = 72$ $\frac{18c}{18} = \frac{72}{18}$ $c = 4$	6. $\frac{r}{6} = 36$ $\frac{r}{6}(6) = 36(6)$ $r = 216$



If students struggle to solve equations algebraically, provide them with concrete manipulatives or encourage them to draw visual models. This will provide additional kinesthetic and visual entry points as students work to solve independently.

12.12.5: Wrap-Up: Error Analysis

Component Overview: Students explain and correct an error in a sample of student work. Use data from this formative assessment to determine which students need remediation before the next lesson.



12.12.5 Wrap-Up: Error Analysis

1. While checking her homework from last night, Lola noticed she got the following problem incorrect. Examine Lola's work and write a note to her detailing how she can correct her mistake.

$$\begin{array}{r} 42 = h \div 7 \\ +7 \quad +7 \\ \hline 49 = h \end{array}$$

*Answers will vary. Lola added 7 to both sides of the equation instead of multiplying. The inverse of division is multiplication. If Lola would have multiplied both sides by 7 she would have gotten the problem correct.
 $h = 294$*



Consider providing students with sentence stems or answer choices if they struggle to begin writing their responses. This activity will help students avoid making similar errors themselves in the future.